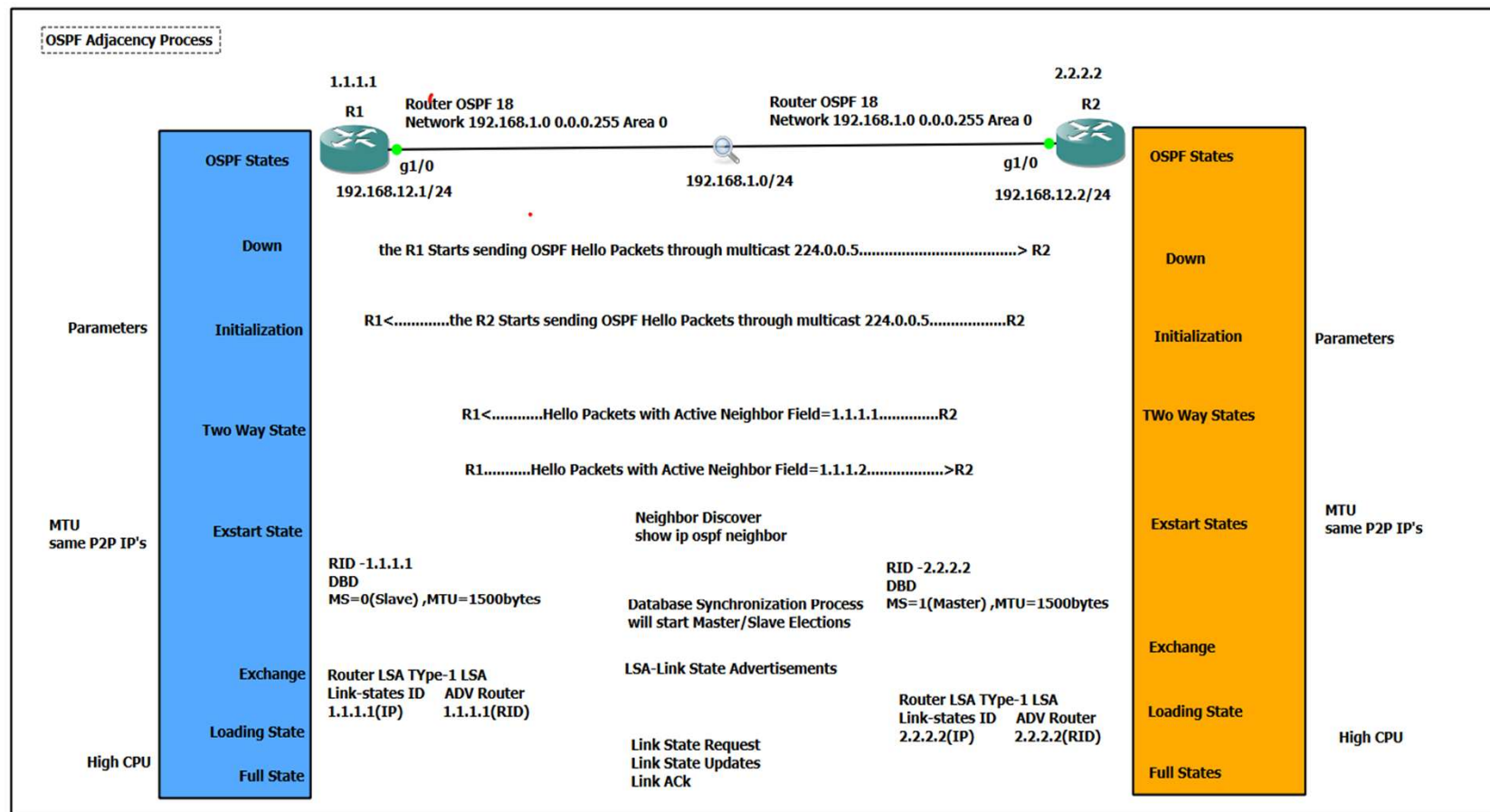




Capturing from Standard input [R1 GigabitEthernet1/0 to R2 GigabitEthernet1/0]

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
84	281.834348	ca:02:43:7c:00:1c	ca:01:73:54:00:1c	ARP	60	192.168.12.2 is at ca:02:43:7c:00:1c
85	282.470037	ca:01:73:54:00:1c	ca:01:73:54:00:1c	LOOP	60	Reply
86	284.163479	192.168.12.1	224.0.0.5	OSPF	94	Hello Packet
87	284.170375	192.168.12.2	192.168.12.1	OSPF	78	DB Description

Wireshark - Packet 104 - Standard input

.0.. = (O) Opaque: Not set
..0. = (DC) Demand Circuits: Not supported
...1 = (L) LLS Data block: Present
.... 0... = (N) NSSA: Not supported
.... .0.. = (MC) Multicast: Not capable
.... ..1. = (E) External Routing: Capable
.... ...0 = (MT) Multi-Topology Routing: No

Router Priority: 1
Router Dead Interval [sec]: 40
Designated Router: 192.168.12.1
Backup Designated Router: 192.168.12.2
Active Neighbor: 1.1.1.1

▼ OSPF LLS Data Block

Checksum: 0xffff6
LLS Data Length: 12 bytes

▼ Extended options TLV

TLV Type: 1
TLV Length: 4

▼ Options: 0x00000001, (LR) LSDB Resynchronization

.... = (RS) Restart Signal: Not set
.... = (LR) LSDB Resynchronization: Set

0050 01 01 ff f6 00 03 00 01 00 04 00 00 00 01

(LR) LSDB Resynchronization (ospf.lls.ext.options.lr), 1 bit

☒ Show packet bytes Layout: Vertical (Stacked) Close Help

Wireshark - Packet 106 - Standard input

Instance ID: Base IPv4 Unicast Instance (0)
Auth Type: Null (0)
Auth Data (none): 0000000000000000

▼ OSPF Hello Packet

Network Mask: 255.255.255.0
Hello Interval [sec]: 10

▼ Options: 0x12, (L) LLS Data block, (E) External Routing

0... = DN: Not set
.0.. = (O) Opaque: Not set
..0. = (DC) Demand Circuits: Not supported
...1 = (L) LLS Data block: Present
.... 0... = (N) NSSA: Not supported
.... .0.. = (MC) Multicast: Not capable
.... ..1. = (E) External Routing: Capable
.... ...0 = (MT) Multi-Topology Routing: No

Router Priority: 1
Router Dead Interval [sec]: 40
Designated Router: 192.168.12.1
Backup Designated Router: 192.168.12.2
Active Neighbor: 2.2.2.2

▼ OSPF LLS Data Block

Checksum: 0xffff6

No.: 106 · Time: 293.222524 · Source: 192.168.12.1 · Destination: 224.0.0.5 · Protocol: OSPF · Length: 94 · Info: Hello Packet

☒ Show packet bytes Layout: Vertical (Stacked) Close Help

Capturing from Standard input [R1 GigabitEthernet1/0 to R2 GigabitEthernet1/0]

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
74	255.866633	ca:02:43:7c:00:1c	ca:02:43:7c:00:1c	LOOP	60	Reply
75	262.468599	ca:01:73:54:00:1c	ca:01:73:54:00:1c	LOOP	60	Reply
76	264.795370	192.168.12.1	224.0.0.5	OSPF	90	Hello Packet

Wireshark · Packet 87 · Standard input

```

0... .. = DN: Not set
.1.. ... = (O) Opaque: Set
..0. ... = (DC) Demand Circuits: Not supported
...1 ... = (L) LLS Data block: Present
... 0... = (N) NSSA: Not supported
... .0.. = (MC) Multicast: Not capable
... ..1 = (E) External Routing: Capable
... ...0 = (MT) Multi-Topology Routing: No
  DB Description: 0x07, (I) Init, (M) More, (MS) Master
    ... 0... = (R) OOBResync: Not set
    ... .1.. = (I) Init: Set
    ... ..1 = (M) More: Set
    ... ...1 = (MS) Master: Yes
  DD Sequence: 8091
  OSPF LLS Data Block
    Checksum: 0xffff6
    LLS Data Length: 12 bytes
  Extended options TLV
    TLV Type: 1
    TLV Length: 4
    Options: 0x00000001, (LR) LSDB Resynchronization
      ... ..0. = (RS) Restart Signal: Not set
  
```

(O) Opaque (ospfv2.options.o), 1 bit

☒ Show packet bytes Layout: Vertical (Stacked) Close Help

The Process of Master/Slave Elections

Wireshark · Packet 89 · Standard input

```

Auth Data (none): 0000000000000000
  OSPF DB Description
    Interface MTU: 1500
    Options: 0x52, (O) Opaque, (L) LLS Data block, (E) External Routing
      0... .. = DN: Not set
      .1.. ... = (O) Opaque: Set
      ..0. ... = (DC) Demand Circuits: Not supported
      ...1 ... = (L) LLS Data block: Present
      ... 0... = (N) NSSA: Not supported
      ... .0.. = (MC) Multicast: Not capable
      ... ..1 = (E) External Routing: Capable
      ... ...0 = (MT) Multi-Topology Routing: No
    DB Description: 0x07, (I) Init, (M) More, (MS) Master
      ... 0... = (R) OOBResync: Not set
      ... .1.. = (I) Init: Set
      ... ..1 = (M) More: Set
      ... ...1 = (MS) Master: Yes
    DD Sequence: 7908
  OSPF LLS Data Block
    Checksum: 0xffff6
    LLS Data Length: 12 bytes
  Extended options TLV
  
```

No.: 89 · Time: 284.195437 · Source: 192.168.12.1 · Destination: 192.168.12.2 · Protocol: OSPF · Length: 78 · Info: DB Description

☒ Show packet bytes Layout: Vertical (Stacked) Close Help

108	295.884402	ca:02:43:7c:00:1c	ca:02:43:7c:00:1c	LOOP	60	Reply
-----	------------	-------------------	-------------------	------	----	-------



No.	Time	Source	Destination	Protocol	Length	Info
74	255.866633	ca:02:43:7c:00:1c	ca:02:43:7c:00:1c	LOOP	60	Reply
75	262.468599	ca:01:73:54:00:1c	ca:01:73:54:00:1c	LOOP	60	Reply
76	264.785370	192.168.12.1	192.168.12.2	OSPF	98	Hello Packet

Wireshark - Packet 90 - Standard input

[Stream index: 8]

- > Internet Protocol Version 4, Src: 192.168.12.1, Dst: 192.168.12.2
- ▼ Open Shortest Path First
 - > OSPF Header
 - ▼ OSPF DB Description
 - Interface MTU: 1500
 - > Options: 0x52, (O) Opaque, (L) LLS Data block, (E) External Routing
 - ▼ DB Description: 0x02, (M) More
 - 0... = (R) OOBResync: Not set
 -0.. = (I) Init: Not set
 -1. = (M) More: Set
 -0 = (MS) Master: No
 - DD Sequence: 8091
 - ▼ LSA-type 1 (Router-LSA), len 48
 - .000 0001 1111 1111 = LS Age (seconds): 511
 - 0... = Do Not Age Flag: 0
 - > Options: 0x22, (DC) Demand Circuits, (E) External Routing
 - LS Type: Router-LSA (1)
 - Link State ID: 1.1.1.1
 - Advertising Router: 1.1.1.1
 - Sequence Number: 0x80000002
 - Checksum: 0xd4c9
 - Length: 48
 - ▼ OSPF LLS Data Block
 - Checksum: 0xffff6

No.: 90 · Time: 284.195437 · Source: 192.168.12.1 · Destination: 192.168.12.2 · Protocol: OSPF · Length: 98 · Info: DB Description

☒ Show packet bytes Layout: Vertical (Stacked) Close Help

Wireshark - Packet 91 - Standard input

- Area ID: 0.0.0.0 (Backbone)
- Checksum: 0x2d28 [correct]
- Instance ID: Base IPv4 Unicast Instance (0)
- Auth Type: Null (0)
- Auth Data (none): 0000000000000000
- ▼ OSPF DB Description
 - Interface MTU: 1500
 - > Options: 0x52, (O) Opaque, (L) LLS Data block, (E) External Routing
 - ▼ DB Description: 0x01, (MS) Master
 - 0... = (R) OOBResync: Not set
 -0.. = (I) Init: Not set
 -0. = (M) More: Not set
 -1 = (MS) Master: Yes
 - DD Sequence: 8092
- ▼ LSA-type 1 (Router-LSA), len 48
 - .000 0000 0000 0001 = LS Age (seconds): 1
 - 0... = Do Not Age Flag: 0
 - > Options: 0x22, (DC) Demand Circuits, (E) External Routing
 - LS Type: Router-LSA (1)
 - Link State ID: 2.2.2.2
 - Advertising Router: 2.2.2.2
 - Sequence Number: 0x80000002
 - Checksum: 0xaae7
 - Length: 48
- > OSPF LLS Data Block

No.: 91 · Time: 284.211375 · Source: 192.168.12.2 · Destination: 192.168.12.1 · Protocol: OSPF · Length: 98 · Info: DB Description

☒ Show packet bytes Layout: Vertical (Stacked) Close Help

Capturing from Standard input [R1 GigabitEthernet1/0 to R2 GigabitEthernet1/0]						
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help						
Apply a display filter ... <Ctrl-/>						
No.	Time	Source	Destination	Protocol	Length	Info
74	255.866633	ca:02:43:7c:00:1c	ca:02:43:7c:00:1c	LOOP	60	Reply
75	262.468599	ca:01:73:54:00:1c	ca:01:73:54:00:1c	LOOP	60	Reply
76	264.795370	192.168.12.1	224.0.0.5	OSPF	90	Hello Packet
77	265.862762	ca:02:43:7c:00:1c	ca:02:43:7c:00:1c	LOOP	60	Reply
78	269.227189	ca:01:73:54:00:1c	CDP/VTP/DTP/PagP/UD...	CDP	369	Device ID: R1 Port ID: GigabitEthernet1/0
79	272.472307	ca:01:73:54:00:1c	ca:01:73:54:00:1c	LOOP	60	Reply
80	274.317747	192.168.12.1	224.0.0.5	OSPF	90	Hello Packet
81	275.877902	ca:02:43:7c:00:1c	ca:02:43:7c:00:1c	LOOP	60	Reply
82	281.802294	192.168.12.2	224.0.0.5	OSPF	90	Hello Packet
83	281.818350	ca:01:73:54:00:1c	Broadcast	ARP	60	Who has 192.168.12.2? Tell 192.168.12.1
84	281.834348	ca:02:43:7c:00:1c	ca:01:73:54:00:1c	ARP	60	192.168.12.2 is at ca:02:43:7c:00:1c
85	282.470037	ca:01:73:54:00:1c	ca:01:73:54:00:1c	LOOP	60	Reply
86	284.163479	192.168.12.1	224.0.0.5	OSPF	94	Hello Packet
87	284.179375	192.168.12.2	192.168.12.1	OSPF	78	DB Description
88	284.179375	192.168.12.2	192.168.12.1	OSPF	94	Hello Packet
89	284.195437	192.168.12.1	192.168.12.2	OSPF	78	DB Description
90	284.195437	192.168.12.1	192.168.12.2	OSPF	98	DB Description
91	284.211375	192.168.12.2	192.168.12.1	OSPF	98	DB Description
92	284.227375	192.168.12.1	192.168.12.2	OSPF	70	LS Request
93	284.227375	192.168.12.1	192.168.12.2	OSPF	78	DB Description
94	284.243376	192.168.12.2	192.168.12.1	OSPF	110	LS Update
95	284.243376	192.168.12.2	192.168.12.1	OSPF	70	LS Request
96	284.259375	192.168.12.1	192.168.12.2	OSPF	110	LS Update
97	284.706394	192.168.12.1	224.0.0.5	OSPF	110	LS Update
98	284.770381	192.168.12.1	224.0.0.5	OSPF	94	LS Update
99	285.883632	ca:02:43:7c:00:1c	ca:02:43:7c:00:1c	LOOP	60	Reply
100	286.761524	192.168.12.1	224.0.0.5	OSPF	78	LS Acknowledge
101	286.777523	192.168.12.2	224.0.0.5	OSPF	118	LS Acknowledge
102	287.302566	192.168.12.2	224.0.0.5	OSPF	110	LS Update
103	289.812661	192.168.12.1	224.0.0.5	OSPF	78	LS Acknowledge
104	291.158792	192.168.12.2	224.0.0.5	OSPF	94	Hello Packet
105	292.461701	ca:01:73:54:00:1c	ca:01:73:54:00:1c	LOOP	60	Reply
106	293.222524	192.168.12.1	224.0.0.5	OSPF	94	Hello Packet
107	295.868417	ca:02:43:7c:00:1c	CDP/VTP/DTP/PagP/UD...	CDP	369	Device ID: R2 Port ID: GigabitEthernet1/0
108	295.884402	ca:02:43:7c:00:1c	ca:02:43:7c:00:1c	LOOP	60	Reply

```
R1 x R2 + - □ x
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
+ - replicated route, % - next hop override

Gateway of last resort is not set

1.0.0.0/32 is subnetted, 1 subnets
  1.1.1.1 is directly connected, Loopback0
192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
  192.168.12.0/24 is directly connected, GigabitEthernet1/0
  192.168.12.1/32 is directly connected, GigabitEthernet1/0

R1#
Sep  4 16:04:18.895: %OSPF-5-ADJCHG: Process 18, Nbr 2.2.2.2 on GigabitEthernet
1/0 from LOADING to FULL, Loading Done
R1#show ip ospf database

        OSPF Router with ID (1.1.1.1) (Process ID 18)

          Router Link States (Area 0)

Link ID      ADV Router    Age      Seq#       Checksum Link count
1.1.1.1      1.1.1.1       703      0x80000003 0x0036F0 2
2.2.2.2      2.2.2.2       701      0x80000003 0x0026F3 2

          Net Link States (Area 0)

Link ID      ADV Router    Age      Seq#       Checksum
192.168.12.1 1.1.1.1       703      0x80000001 0x00C7EB
R1#
```

```
R1 R2 + -
Overwrite the previous NVRAM configuration?[confirm]
Building configuration...
[OK]
R2(config-router)#end
R2#
*Sep  4 16:05:44.291: %SYS-5-CONFIG_I: Configured from console by console
R2#show running-config | s ospf
router ospf 18
  network 2.2.2.2 0.0.0.0 area 0
  network 192.168.12.0 0.0.0.255 area 0
R2#show ip ospf database
                        ^
% Invalid input detected at '^' marker.

R2#show ip ospf database

        OSPF Router with ID (2.2.2.2) (Process ID 18)

          Router Link States (Area 0)

Link ID      ADV Router    Age      Seq#       Checksum Link count
1.1.1.1      1.1.1.1       730      0x80000003 0x0036F0 2
2.2.2.2      2.2.2.2       726      0x80000003 0x0026F3 2

          Net Link States (Area 0)

Link ID      ADV Router    Age      Seq#       Checksum
192.168.12.1 1.1.1.1       729      0x80000001 0x00C7EB
R2#
```